

Microfoundations IV: General Equilibrium

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Outline

- 1 Introduction
- 2 2-Period Economy
- 3 The First Welfare Theorem
- 4 General Equilibrium in an Infinite-Period Economy



General Equilibrium Analysis

- We now integrate the many individual decisions studied separately:
 - ▶ consumption and savings
 - ▶ consumption and leisure
 - ▶ investment
- We study under which conditions we have equilibrium in all the different markets
- To do so, we need some critical simplifications:
 - ▶ a **representative household** making consumption-savings-leisure decisions
 - ▶ a **representative firm** choosing to hire labor and rent capital
 - ▶ a **representative investment firm** carrying out investment
- We start with the case of a 2-period economy to then extend this analysis



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GE in 2-period economy

- The economy starts with a given level of capital, K_1
- Production function: $F(K, L)$
- Depreciation rate δ for K_1 ; capital in period 2 fully depreciates

(1) Representative Household:

$$\text{Max}_{c_1, l_1, c_2, l_2} u(c_1) + v(l_1) + \beta [u(c_2) + v(l_2)]$$

$$\text{s.t.: } c_1 + \frac{c_2}{1+r} \leq w_1(1-l_1) + \frac{w_2(1-l_2)}{1+r} + K_1(1+r_1^K - \delta) + \Pi$$

where: $\Pi \equiv \Pi_1^F + \frac{\Pi_2^F}{1+r} + \Pi'$ ($\Pi = 0$ in equilibrium)

F.O.C.:

$$\frac{v'(l_t)}{u'(c_t)} = w_t \quad (1)$$

$$u'(c_1) = \beta(1+r)u'(c_2) \quad (2)$$



GE in 2-period economy

(2) Representative Firm:

$$\Pi_t^F = \max_{K,L} F(K_t, L_t) - w_t L_t - r_t^K K_t$$

F.O.C.:

$$F_K(K_t, L_t) - r_t^K = 0 \quad (3)$$

$$F_L(K_t, L_t) - w_t = 0 \quad (4)$$

(3) Representative Investment Firm:

$$\Pi' = \max_l \frac{r_2^K [(1 - \delta)K_1 + l]}{1 + r} - [(1 - \delta)K_1 + l]$$

F.O.C.:

$$1 + r = r_2^K \quad (5)$$



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The First Welfare Theorem

- Is the decentralized, competitive equilibrium allocation just described **Pareto optimal**?
 - ▶ Pareto Optimality (Efficiency): there are no ways to make one person better-off without making some other person worse-off (there are no possibilities for Pareto improvements)
 - ▶ Vilfredo Pareto (1848-1923): *Manual of Political Economy* (1906)
 - ▶ Allocation optimality/efficiency doesn't guarantee fair distribution/equity
- Let's invoke the metaphor of a benevolent "social planner"
- The fictitious, benevolent social planner solves, for given K_1 :

$$\text{Max}_{c_1, l_1, c_2, l_2, L_1, L_2, K_2} u(c_1) + v(l_1) + \beta [u(c_2) + v(l_2)]$$

s.t.:

$$K_2 \leq (1 - \delta)K_1 + F(K_1, L_1) - c_1$$

$$c_2 \leq F(K_2, L_2)$$

$$L_t \leq 1 - l_t \text{ for } t = 1, 2$$



The First Welfare Theorem

- Thus obtaining the objective function:

$$l_t = 1 - L_t$$

$$u((1-\delta)K_1 + F(K_1, L_1) - K_2) + v(1-L_1) + \beta[u(F(K_2, L_2)) + v(1-L_2)]$$

- $$\begin{aligned} u'((1-\delta)K_1 + F(K_1, L_1) - K_2)F_L(K_1, L_1) - v'(1-L_1) &= 0 \\ \beta[u'(F(K_2, L_2))F_L(K_2, L_2) - v'(1-L_2)] &= 0 \\ -u'((1-\delta)K_1 + F(K_1, L_1) - K_2) + \beta[u'(F(K_2, L_2))F_K(K_2, L_2)] &= 0 \end{aligned}$$

which can be arranged to get back to the competitive equilibrium equations (10) and (11)



The First Welfare Theorem

If an allocation $\{c_1, l_1, c_2, l_2, l, K_2, L_1, L_2\}$ is part of a competitive equilibrium, then it solves the social planner's problem

- Proceed by contradiction: the allocation $\{c_1, l_1, c_2, l_2, l, K_2, L_1, L_2\}$ with prices $\{w_1, w_2, r_1^K, r_2^K, r\}$ is a competitive equilibrium, but there is another allocation $\{\hat{c}_1, \hat{l}_1, \hat{c}_2, \hat{l}_2, \hat{l}, \hat{K}_2, \hat{L}_1, \hat{L}_2\}$ that is Pareto optimal (i.e., satisfies the constraints on the planner's problem but yield a strictly higher utility level).
- This implies that the Pareto optimal allocation is not affordable to the household at competitive equilibrium prices:

$$\hat{c}_1 + \frac{\hat{c}_2}{1+r} > w_1(1 - \hat{h}_1) + \frac{w_2(1 - \hat{l}_2)}{1+r} + K_1(1 + r_1^K - \delta) + \Pi \quad (12)$$

(continues)



The First Welfare Theorem

Proof.

- It also implies that profit-maximizing firms make at least as much profit as they would by choosing the alternative quantities:

$$\Pi_1^F = F(K_1, L_1) - w_1 L_1 - r_1^K K_1 \geq F(K_1, \hat{L}_1) - w_1 \hat{L}_1 - r_1^K K_1$$

$$\Pi_2^F = F(K_2, L_2) - w_2 L_2 - r_2^K K_2 \geq F(\hat{K}_2, \hat{L}_2) - w_2 \hat{L}_2 - r_2^K \hat{K}_2$$

$$\Pi' = \left(\frac{r_2^K}{1+r} - 1 \right) [(1-\delta)K_1 + I] \geq \left(\frac{r_2^K}{1+r} - 1 \right) [(1-\delta)K_1 + \hat{I}]$$

- Given that $\Pi \equiv \Pi_1^F + \frac{\Pi_2^F}{1+r} + \Pi'$, we get that:

$$\begin{aligned} \Pi \geq & F(K_1, \hat{L}_1) - w_1 \hat{L}_1 - r_1^K K_1 + \frac{F(\hat{K}_2, \hat{L}_2) - w_2 \hat{L}_2 - r_2^K \hat{K}_2}{1+r} \\ & + \left(\frac{r_2^K}{1+r} - 1 \right) \left[(1-\delta)K_1 + \hat{I} \right] \end{aligned} \quad (13)$$

(continues)



The First Welfare Theorem

Proof.

- Use (13), $\hat{L}_t \leq 1 - \hat{l}_t$, and $\hat{I} = \hat{K}_2 - (1 - \delta)K_1$ into (12):

$$\begin{aligned} \hat{c}_1 + \frac{\hat{c}_2}{1+r} &> w_1 \hat{L}_1 + \frac{w_2 \hat{L}_2}{1+r} + K_1(1+r_1^K - \delta) + F(K_1, \hat{L}_1) - w_1 \hat{L}_1 - r_1^K K_1 + \\ &+ \frac{F(\hat{K}_2, \hat{L}_2) - w_2 \hat{L}_2 - r_2^K \hat{K}_2}{1+r} + \left(\frac{r_2^K}{1+r} - 1 \right) [(1-\delta)K_1 + \\ &+ \hat{K}_2 - (1-\delta)K_1] = (1-\delta)K_1 + F(K_1, \hat{L}_1) + \frac{F(\hat{K}_2, \hat{L}_2)}{1+r} - \hat{K}_2 \end{aligned}$$

- Thus:

$$\hat{c}_1 + \hat{K}_2 + \frac{\hat{c}_2}{1+r} > (1-\delta)K_1 + F(K_1, \hat{L}_1) + \frac{F(\hat{K}_2, \hat{L}_2)}{1+r}$$

- Which in turn implies that at least one of the following is true (contradicting planner's constraints):

$$\hat{c}_1 + \hat{K}_2 > (1 - \delta)K_1 + F(K_1, \hat{L}_1)$$

$$\hat{c}_2 > F(\hat{K}_2, \hat{L}_2)$$



The First Welfare Theorem

- The social planner maximizes household's utility subject to technological possibilities
- A competitive equilibrium has:
 - ▶ Household maximizing utility subject to prices
 - ▶ Prices reflecting technological possibilities
- So in a competitive equilibrium household is indirectly maximizing utility subject to technological constraints
- The First Welfare Theorem also holds:
 - ▶ For differentiated goods at each date
 - ▶ For exogenous technological progress
 - ▶ In an economy with uncertainty and complete markets
 - ▶ For differentiated households (different preferences, different abilities, different wealth). The challenge is to properly define the social planner's preferences
- This Theorem is very useful for thinking about public policy: efficiency vs. equity



The First Welfare Theorem

- The First Welfare Theorem fails when there is:
 - ▶ Monopoly power
 - ▶ Externalities
 - ▶ Asymmetric information
 - ▶ Incomplete markets
 - ▶ Borrowing constraints
- Starting point of public policy analysis is asking what is the failure of the First Welfare Theorem



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Infinite-Period Economy

Representative Household:

$$\text{Max}_{c_t, l_t, a_{t+1}} \sum_{t=0}^{\infty} \beta^t [u(c_t) + v(l_t)]$$

s.t.:

$$\begin{aligned} a_{t+1} &= (1 + r_t)a_t + w_t(1 - l_t) + \Pi_t^F + \Pi_t^I - c_t \\ a_0 &= K_0(1 + r_0^K - \delta) \text{ given} \\ \lim_{T \rightarrow \infty} \frac{a_T}{(1 + r)^T} &\geq 0 \end{aligned}$$

Representative Firm: maximizes profit period-by-period, as before

Representative Investment Firm: only difference is that capital now doesn't fully depreciates

$$\Pi'_t = \max_l \frac{r_{t+1}^K + 1 - \delta}{1 + r_{t+1}} [(1 - \delta)K_t + I_t] - [(1 - \delta)K_t + I_t]$$



Infinite-Period Economy

Definition

A competitive equilibrium consists of:

- 1 An allocation $\{c_t, l_t, I_t, K_{t+1}, L_t\}_{t=0}^{\infty}$
- 2 Prices $\{w_t, r_t^K, r_t\}_{t=0}^{\infty}$

such that:

- 1 $\{c_t, l_t\}_{t=0}^{\infty}$ solves the household's problem, for given prices
- 2 L_t, K_t solve the firm's problem for every t , taking prices as given
- 3 I_t solves the investment firm's problem for every t , taking prices as given
- 4 Goods and Labor markets clear for every t :
 - a Goods: $F(K_t, L_t) = c_t + I_t$
 - b Capital: $K_{t+1} = K_t(1 - \delta) + I_t$
 - c Labor: $L_t + l_t = 1$

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Infinite-Period Economy

- No conceptual difference to the two-period model
- Similar F.O.C.:

$$\frac{v'(l_t)}{u'(c_t)} = w_t \quad (14)$$

$$u'(c_t) = \beta(1 + r_{t+1})u'(c_{t+1}) \quad (15)$$

$$F_K(K_t, L_t) - r_t^K = 0 \quad (16)$$

$$F_L(K_t, L_t) - w_t = 0 \quad (17)$$

$$r_{t+1} = r_{t+1}^K - \delta \quad (18)$$

- From which we get similar equilibrium conditions:

$$\frac{v'(l_t)}{u'(c_t)} = F_L(K_t, L_t) \quad (19)$$

$$\frac{u'(c_t)}{\beta u'(c_{t+1})} = (1 + F_K(K_{t+1}, L_{t+1}) - \delta) \quad (20)$$

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Infinite-Period Economy

- Thus (20) becomes:

- Market clearing:

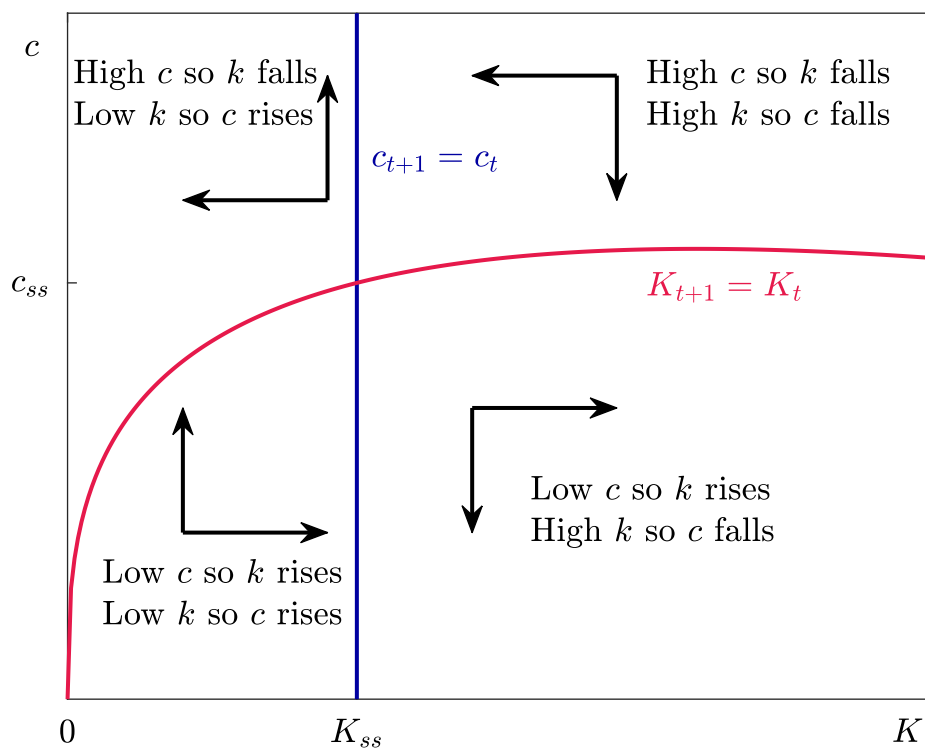
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- In a stationary (i.e. long-run) equilibrium: $c_{t+1} = c_t$ and $K_{t+1} = K_t$
- From (21):

- From (22):

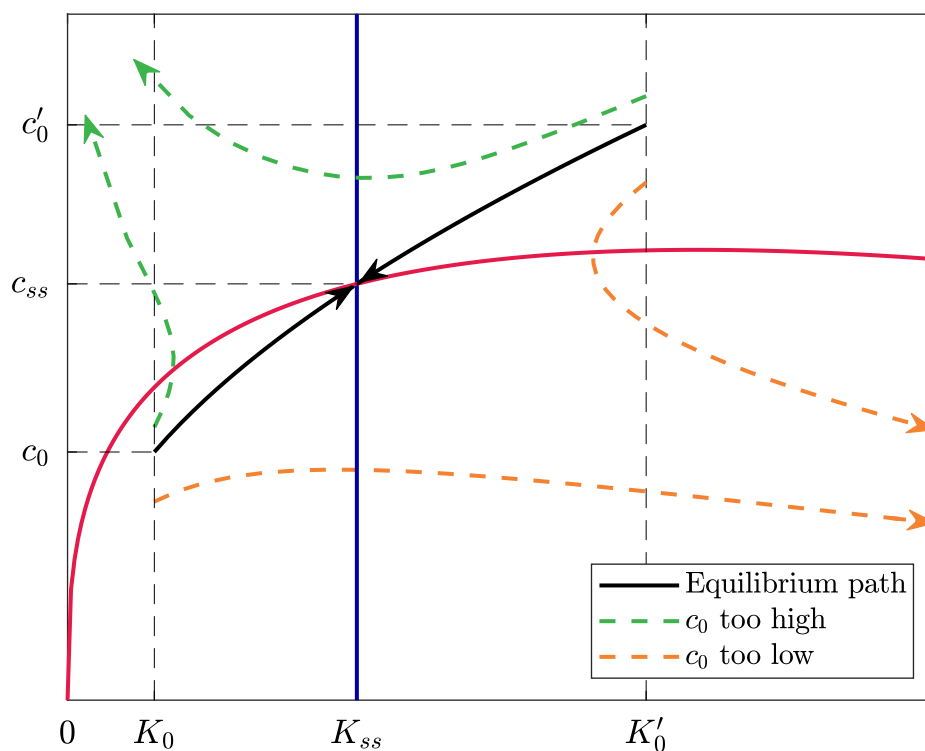
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Infinite-Period Economy



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Infinite-Period Economy



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Infinite-Period Economy: The Golden Rule

- The First Welfare Theorem tells us that the competitive equilibrium is Pareto Optimal but it doesn't maximize steady-state consumption
- Household cares about $\{c_t\}_{t=0}^{\infty}$ and not just steady-state consumption
- **Impatient** household chooses to consume a bit more in the present and accepting lower steady-state consumption level

$$\Rightarrow K_{ss} < K_{gr}$$

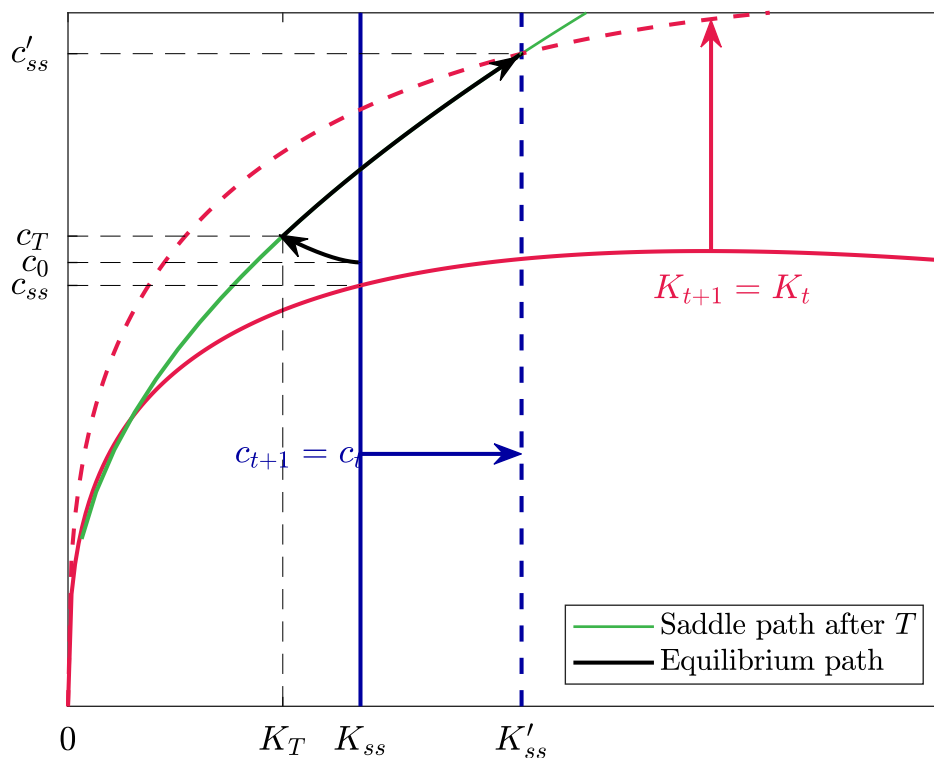


Infinite-Period Economy: Anticipation Effects

- Suppose an economy in steady state when the news of a technological discovery arrives: at T , $F(K, 1)$ becomes $AF(K, 1)$, with $A > 1$
- At T :
 - ▶ the $c_t = c_{t+1}$ schedule shifts to the right: $AF_K(K, 1) = \frac{1}{\beta} - 1 + \delta$
 - ▶ the $K_t = K_{t+1}$ schedule shifts up
 - ▶ the economy has to be in the new saddle path
- At the moment the news arrive, agents can adjust consumption (jump) to put it into an “equilibrium path” (i.e. stock of capital is a state variable and c is the choice variable)
- The initial dynamics governs the economy until T

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Infinite-Period Economy: Anticipation Effects



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